

SUN LAZER

Dual Heater Process Controller

Complete Functional & Technical Documentation

Revision 1.0 | 09 August 2026

1. Project Purpose

This document defines the complete functional and technical specification for a Sun Lazer dual-heater process controller.

The controller shall operate two heaters independently, maintain programmed temperatures for a programmed HH:MM:SS process time, control a bidirectional motor, monitor two travel limit switches and torque, provide a TFT user interface, store ten programs, log process information, and expose a local ESP32 web dashboard.

2. System Architecture

Main controller: ESP32. User interface: SPI TFT LCD and five navigation push buttons. Temperature: two independent industrial temperature sensors with suitable interface modules. Heater outputs: two isolated SSR/contactor control channels. Motor: protected forward/reverse control. Position: DOWN/PROCESS and HOME/ORIGINAL limit switches. Load monitoring: torque sensor. Storage: ESP32 NVS for settings/programs and microSD for long-term history. Communication: USB Serial and Wi-Fi local web server.

3. Operating Sequence

Power ON → logo screen for about 5 seconds → Home screen → select settings/program → safety check → motor moves DOWN → DOWN limit stops motor → heaters reach setpoints → process timer starts → controller maintains both temperatures while monitoring torque and safety → timer completes → heaters enter configured safe state → motor moves UP → HOME limit stops motor → process record is saved → Process Complete → Ready.

4. TFT Screens

Boot: SUN LAZER / INDUSTRIAL SYSTEM / Initializing.

Home: Heater 1 actual/set temperature, Heater 2 actual/set temperature, remaining timer, status, and menu options.

Temperature: independently edit Heater 1 and Heater 2 setpoints and save.

Timer: edit HH:MM:SS and save.

Process: show actual/set temperatures, remaining time, torque, motor state, limit states and status.

Programs: Program 01 through Program 10; each program contains Heater 1 temperature, Heater 2 temperature, process timer, torque limit and temperature tolerance.

Service: sensor test, heater test, limit-switch test and motor jog with safety interlocks.

Settings: calibration, tolerance, torque limit, date/time, Wi-Fi and factory reset.

5. Temperature Control

Each heater must have its own sensor and control loop. PID control is recommended for stable industrial temperature control. The ESP32 shall never drive a heater directly; it shall drive an appropriately rated isolated SSR or contactor interface.

Temperature sensor failure shall place the affected heater in a safe state and generate an alarm. Independent hardware over-temperature protection is recommended and should not depend on the ESP32.

6. Motor and Limit Control

Two travel inputs are required: DOWN_LIMIT for the process position and HOME_LIMIT for the original position. A bidirectional motor requires an appropriate reversing contactor/driver arrangement selected for the actual motor type and power.

During DOWN movement, the motor must stop immediately when DOWN_LIMIT becomes active. During UP movement, the motor must stop immediately when HOME_LIMIT becomes active. Software must also include timeout protection so a failed limit switch cannot leave the motor running indefinitely.

7. Torque Monitoring

The torque sensor shall be selected according to the motor/shaft arrangement and its electrical output (for example 0–5 V, 0–10 V, 4–20 mA or bridge output). The firmware shall display torque, record maximum torque and stop the motor/process when the configured torque limit is exceeded. Sensor scaling and calibration must be stored in settings.

8. Ten Program Recipe System

Each program stores: program name/number, Heater 1 setpoint, Heater 2 setpoint, HH:MM:SS process time, torque limit and temperature tolerance. Selecting a program loads its values into the process controller. Program editing requires an explicit Save action.

9. Data Logging

Store configuration and programs in ESP32 non-volatile storage. For approximately one year of process history, use a microSD card. Recommended record fields: date, time, program, H1 set/actual, H2 set/actual, process time, maximum torque, motor result, alarm code and final status.

10. Serial Log Format

Example events: [BOOT] SUN LAZER CONTROLLER; [PROGRAM] P02 SELECTED; [MOTOR] DOWN START; [LIMIT] DOWN ACTIVE; [PROCESS] HEATING; [PROCESS] TEMPERATURE READY; [PROCESS] TIMER START; [TORQUE] 32.5 Nm; [PROCESS] TIMER COMPLETE; [MOTOR] UP START; [LIMIT] HOME ACTIVE; [PROCESS] COMPLETE.

11. Local Web Dashboard

The ESP32 shall provide a local Wi-Fi network/dashboard. Dashboard pages should include Live Status, Programs, History and Alarms. Live values should include both temperatures, setpoints, remaining time, torque, motor direction/state, limit-switch states, selected program and alarm/status.

12. Safety Requirements

Use a physical emergency stop and suitable electrical safety circuit; do not rely on software alone. Provide correctly rated fuses/MCBs, motor overload protection, heater protection, grounding/earthing and an independent thermal cutoff/thermostat as appropriate. All mains wiring and protective devices must be selected and installed by a qualified person.

On sensor failure, emergency stop, torque overload, motor fault or travel fault, the controller shall place hazardous outputs into a safe state and show an alarm. Motor movement must have travel limits and a maximum movement timeout.

13. Recommended I/O Map

Example only; finalize after selecting exact hardware: TFT MOSI GPIO23, MISO GPIO19, SCLK GPIO18, CS GPIO5, DC GPIO2, RESET GPIO4; buttons UP32, DOWN33, RIGHT25, OK26, LEFT27; Heater1 GPIO16; Heater2 GPIO17; Motor DOWN GPIO21; Motor UP GPIO22; DOWN_LIMIT GPIO34; HOME_LIMIT GPIO35; Torque analog input GPIO36. GPIO34–39 are input-only on classic ESP32 and require external biasing where needed.

14. Firmware Structure

Use a non-blocking state machine. Suggested modules: Config, Pins, Display, Buttons, Temperature, HeaterControl, MotorControl, Torque, Process, Programs, Storage, WebServer, Logger and Safety. Avoid long delay() calls during operation; use elapsed-time logic so safety, sensors, web requests and display remain responsive.

15. Acceptance Test Checklist

Boot/logo test; button test; TFT test; temperature sensor calibration test; heater channel test; program save/load test; timer accuracy test; DOWN limit test; HOME limit test; motor direction test; motor timeout test; torque scaling test; torque overload test; sensor disconnect test; emergency stop test; power-cycle settings retention test; SD logging test; web dashboard test; complete automatic cycle test.

16. Website/Product Documentation Content

Product name: Sun Lazer Dual Heater Process Controller. Suggested website sections: Product Overview, Key Features, Working Sequence, TFT Interface, 10 Program Recipes, Temperature Control, Motor/Limit Safety, Torque Monitoring, Data Logging, Local Web Dashboard, Applications, Technical Specifications, Safety, Support and Downloads.

Do not publish unverified electrical ratings. Heater power, motor power, sensor range, enclosure rating, supply voltage and torque range must be filled from the final selected components and machine design.

17. Quick Reference – Process State Machine

Step	State
1	IDLE
2	SAFETY CHECK
3	MOVE DOWN
4	DOWN LIMIT
5	HEAT TO SETPOINT
6	TEMPERATURE READY
7	PROCESS TIMER
8	TIMER COMPLETE
9	MOVE UP
10	HOME LIMIT
11	SAVE RECORD
12	PROCESS COMPLETE
13	READY

18. Document Control

Document owner: Sun Lazer

Document type: Product / Engineering Specification

Revision: 1.0

Status: Prototype-to-production baseline